

Tommaso Aiello | [linkedin.com/in/tommaso-aiello-a47674255](https://www.linkedin.com/in/tommaso-aiello-a47674255)
+39 347 518 2845 | tommasoaiello0804@gmail.com
Milan, Italy | github.com/AielloTommaso | aiellotommaso.github.io

EDUCATION

Bocconi University **Milan, Italy | Sep 2025 - Jul 2027**

MSc in Artificial Intelligence | Current average: 28.6/30

- Associazione San Paolo merit scholar; one of 20 recipients selected for a full-tuition waiver.
- Selected for an exchange semester at the University of Maryland, Jan - Jun 2027.

Bocconi University **Milan, Italy | Sep 2022 - Jul 2025**

BSc in Applied Mathematics for Artificial Intelligence | Final grade: 108/110

- Bachelor's thesis: Diffusion Processes for Generative Hierarchical Models; awarded the highest mark.
- Exchange semester at City University of Hong Kong, Jan - Jun 2025.
- Selected coursework: Mathematical Statistics (30 cum laude), Mathematical Modelling for Finance (30/30), Stochastic Processes (29/30), and Algorithms for Optimization and Inference (30 cum laude).

PROFESSIONAL EXPERIENCE

BNL BNP Paribas - Data Analyst Intern **Rome | Jul - Sep 2024**

Banker Performance and Portfolio Analytics

- Built a SQL, Python and Excel pipeline to analyze approximately 10,000 open investment positions and hundreds of thousands of transactions across portfolios managed by approximately 1,000 bankers.
- Built short-horizon Monte Carlo forecasts from empirical transition probabilities and predicted year-end Net Promoter Score within 0.3 points, validating outputs with MSE and cross-entropy.
- Built a Power BI and Excel interface for volume, client returns, market-relative performance and asset allocation; presented it to management and secured permanent adoption after testing.

RESEARCH AND PROJECTS

Bocconi University - Master's Research **Milan | Jan 2026 - Present**

Curie-Weiss and Diffusion | Supervisor: Marc Mézard

- Derived closed-form diffusion likelihoods, posterior log-odds, denoising scores and Hessian-based spinodal times for hierarchical Boltzmann mixtures under Ornstein-Uhlenbeck diffusion.
- Validated coarse-to-fine recovery with a 130,000-sample Monte Carlo benchmark and parameter sweeps; 95% crossing errors were 0.027/0.019, with pooled RMSE of 0.038.

BAINSA - Cryptocurrency Market Trend Forecasting **Mar - Sep 2024**

Supervisor: Prof. R. Duran | Python, FinBERT, PyTorch

- Analyzed hourly time series for approximately 100 cryptocurrencies and one million X/Reddit items using pre-trained FinBERT, correlation and OLS; identified significant links with next-period outcomes (p-values approximately 0.03).
- Developed a sentiment-adjusted Black-Scholes implied-volatility framework and trained a 30-million-parameter neural network to predict the correction term.

Diffusion Processes for Generative Hierarchical Models **Python | PyTorch**

- Built a reproducible Transformer-denoiser framework and ran 1,000+ experiments; found phase transitions required sufficient hierarchy and dimensionality, while low-structure regimes memorized or failed to learn.

Pydeia - 1st of 24 teams, InnovAIte Hackathon **Nov 2025**

- In a five-person team, developed the semantic recommendation and data-ingestion pipeline for Italian university programmes; delivered a working Flask and React voice-first agentic-AI demo in 24 hours.

SKILLS AND ADDITIONAL INFORMATION

- **Programming:** Python (PyTorch, NumPy, pandas, scikit-learn), SQL, R, C, C++, MATLAB, JavaScript and Git.
- **Machine learning:** Deep-learning model training, financial NLP, sentiment analysis, transformer inference, time-series modelling, computer vision, two-view geometry and 3D reconstruction.
- **Quantitative and tools:** Stochastic processes, Monte Carlo, Markov chains, OLS, statistical modelling, optimization, model validation, Excel and Power BI.
- **Languages and additional:** Italian (native), English (C1; Cambridge English C1, 2020; IELTS Academic 7.5, 2024), Spanish (conversational); fully funded nine-month exchange in Ottawa; competitive swimming and bass guitar.